

Lecture Notes: Applications of Integrals

Calculus I

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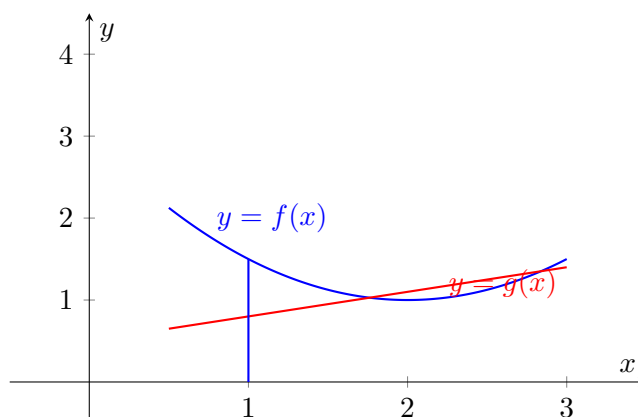
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Section 1 The Area Between Curves

Definition 1 (Area Between Two Curves) If f and g are continuous functions on $[a, b]$ with $f(x) \geq g(x)$ for all $x \in [a, b]$, then the area A of the region bounded by the graphs of f , g , and the vertical lines $x = a$ and $x = b$ is

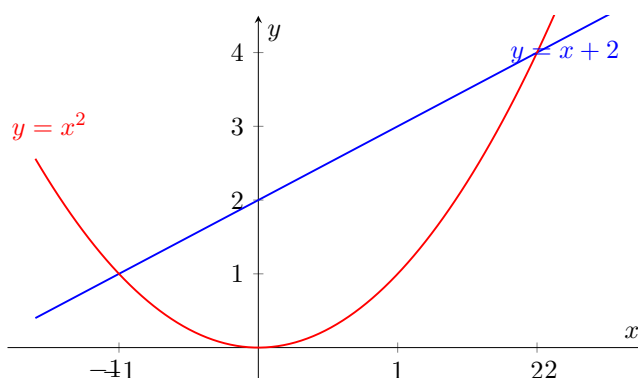
$$A = \int_a^b [f(x) - g(x)] dx.$$

Remark 1 The formula can be interpreted as the limit of Riemann sums: the region is divided into thin vertical strips of width Δx and height $f(x) - g(x)$, so the area of each strip is approximately $[f(x) - g(x)]\Delta x$.



Example 1 Find the area of the region bounded by $y = x^2$ and $y = x + 2$. First, find intersection points: $x^2 = x + 2 \implies x^2 - x - 2 = 0 \implies (x - 2)(x + 1) = 0$. So the curves intersect at $x = -1$ and $x = 2$. On $[-1, 2]$, $x + 2 \geq x^2$. Thus,

$$A = \int_{-1}^2 [(x + 2) - x^2] dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{10}{3} + \frac{7}{6} = \frac{9}{2}.$$



Definition 2 (Area Between Curves with Respect to y) If a region is bounded by curves $x = R(y)$ and $x = L(y)$ with $R(y) \geq L(y)$ on $[c, d]$, then

$$A = \int_c^d [R(y) - L(y)] dy.$$

Example 2 Find the area bounded by $x = y^2$ and $x = y + 2$. Intersection: $y^2 = y + 2 \implies y^2 - y - 2 = 0 \implies (y - 2)(y + 1) = 0$. So $y = -1$ and $y = 2$. On $[-1, 2]$, $y + 2 \geq y^2$. Thus,

$$A = \int_{-1}^2 [(y + 2) - y^2] dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2 = \frac{10}{3} + \frac{7}{6} = \frac{9}{2}.$$

Remark 2 Choosing the correct variable of integration is crucial. If the curves are easier to express as $x = f(y)$, integrate with respect to y .

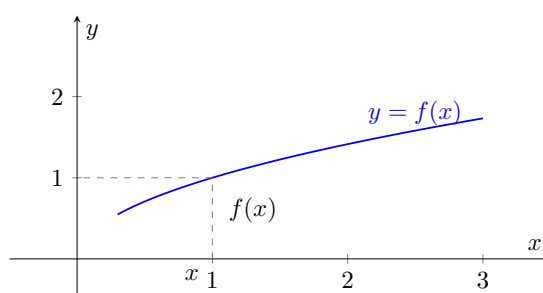
Section 2 Volumes of Solids

Definition 3 (Volume by Slicing) If a solid extends from $x = a$ to $x = b$ and the cross-sectional area perpendicular to the x -axis at x is $A(x)$, then the volume V is

$$V = \int_a^b A(x) dx.$$

Definition 4 (Disk Method) If a region under $y = f(x) \geq 0$ on $[a, b]$ is revolved about the x -axis, the volume of the solid of revolution is

$$V = \int_a^b \pi [f(x)]^2 dx.$$

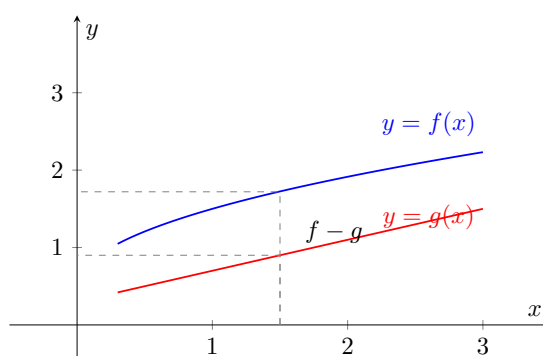


Example 3 Find the volume of the solid obtained by revolving the region bounded by $y = \sqrt{x}$, $x = 1$, $x = 4$, and $y = 0$ about the x -axis.

$$V = \int_1^4 \pi (\sqrt{x})^2 dx = \pi \int_1^4 x dx = \pi \left[\frac{x^2}{2} \right]_1^4 = \pi \left(8 - \frac{1}{2} \right) = \frac{15\pi}{2}.$$

Definition 5 (Washer Method) If the region between $y = f(x)$ and $y = g(x)$ (with $f(x) \geq g(x) \geq 0$) on $[a, b]$ is revolved about the x -axis, the volume is

$$V = \int_a^b \pi ([f(x)]^2 - [g(x)]^2) dx.$$



Example 4 Find the volume of the solid obtained by revolving the region bounded by $y = x$ and $y = x^2$ about the x -axis. Intersection: $x = x^2 \implies x(1 - x) = 0 \implies x = 0, 1$. On $(0, 1)$, $x \geq x^2$. Thus,

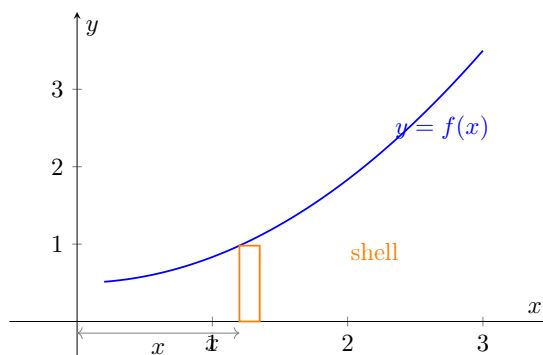
$$V = \int_0^1 \pi (x^2 - x^4) dx = \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}.$$

Definition 6 (Cylindrical Shells Method) If a region under $y = f(x) \geq 0$ on $[a, b]$ with $0 \leq a < b$ is revolved about the y -axis, the volume is

$$V = \int_a^b 2\pi x f(x) dx.$$

More generally, if the region between $y = f(x)$ and $y = g(x)$ (with $f(x) \geq g(x)$) is revolved about the y -axis, then

$$V = \int_a^b 2\pi x [f(x) - g(x)] dx.$$



Example 5 Find the volume of the solid obtained by revolving the region bounded by $y = x - x^2$ and $y = 0$ about the y -axis. The curve intersects the x -axis at $x = 0$ and $x = 1$. Using cylindrical shells,

$$V = \int_0^1 2\pi x(x - x^2) dx = 2\pi \int_0^1 (x^2 - x^3) dx = 2\pi \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 2\pi \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{\pi}{6}.$$

Theorem 1 (Volume Summary) • Disk (about x -axis): $V = \pi \int_a^b [f(x)]^2 dx$.

- Washer (about x -axis): $V = \pi \int_a^b ([f(x)]^2 - [g(x)]^2) dx$.
- Shells (about y -axis): $V = 2\pi \int_a^b x f(x) dx$.
- Shells (general): $V = 2\pi \int_a^b (\text{radius})(\text{height}) dx$.

Section 3 Arc Length

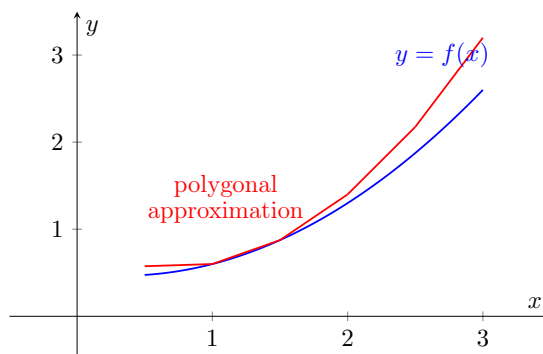
Definition 7 (Arc Length) If f' is continuous on $[a, b]$, then the length L of the curve $y = f(x)$ from $x = a$ to $x = b$ is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

Divide $[a, b]$ into n subintervals. The length of the segment between $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$ is

$$\sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{(\Delta x)^2 + (f'(c_i)\Delta x)^2} = \sqrt{1 + [f'(c_i)]^2} \Delta x$$

by the Mean Value Theorem. Summing and taking the limit as $n \rightarrow \infty$ gives the integral.



Example 6 Find the length of the curve $y = x^{3/2}$ from $x = 0$ to $x = 4$. $f'(x) = \frac{3}{2}\sqrt{x}$, so $[f'(x)]^2 = \frac{9}{4}x$. Thus,

$$L = \int_0^4 \sqrt{1 + \frac{9}{4}x} dx.$$

Let $u = 1 + \frac{9}{4}x$, $du = \frac{9}{4}dx$, so $dx = \frac{4}{9}du$. Limits: $x = 0 \rightarrow u = 1$, $x = 4 \rightarrow u = 10$.

$$L = \frac{4}{9} \int_1^{10} \sqrt{u} du = \frac{4}{9} \cdot \frac{2}{3} [u^{3/2}]_1^{10} = \frac{8}{27} (10\sqrt{10} - 1) \approx 9.07.$$

Definition 8 (Arc Length for Parametric Curves) If a curve is given parametrically by $x = f(t)$, $y = g(t)$ for $t \in [\alpha, \beta]$, where f' and g' are continuous, then the arc length is

$$L = \int_{\alpha}^{\beta} \sqrt{[f'(t)]^2 + [g'(t)]^2} dt = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

Example 7 Find the length of the circle of radius r parametrized by $x = r \cos t$, $y = r \sin t$ for $t \in [0, 2\pi]$. $\frac{dx}{dt} = -r \sin t$, $\frac{dy}{dt} = r \cos t$. Thus,

$$L = \int_0^{2\pi} \sqrt{r^2 \sin^2 t + r^2 \cos^2 t} dt = \int_0^{2\pi} r dt = 2\pi r.$$

Definition 9 (Arc Length of a Function of y) If $x = g(y)$ with g' continuous on $[c, d]$, then

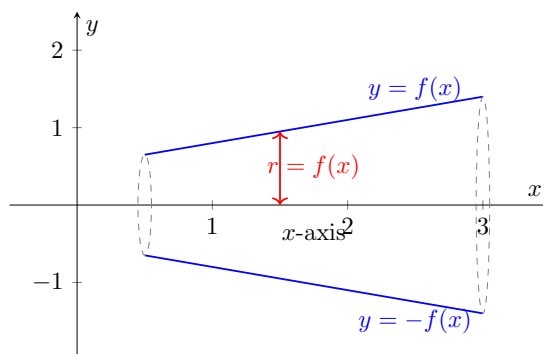
$$L = \int_c^d \sqrt{1 + [g'(y)]^2} dy.$$

Section 4 Areas of Surfaces of Revolution

Definition 10 (Surface Area about the x -Axis) If f is positive and has a continuous derivative on $[a, b]$, the area S of the surface generated by revolving the curve $y = f(x)$ about the x -axis is

$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx.$$

Consider a small segment of arc length $\Delta s \approx \sqrt{1 + [f'(x)]^2} \Delta x$. When revolved about the x -axis, it generates a frustum of a cone with lateral area approximately $2\pi f(x) \Delta s$. Integrating gives the formula.



Example 8 Find the area of the surface generated by revolving $y = \sqrt{x}$ from $x = 1$ to $x = 4$ about the x -axis. $f'(x) = \frac{1}{2\sqrt{x}}$, so $1 + [f'(x)]^2 = 1 + \frac{1}{4x} = \frac{4x+1}{4x}$. Thus,

$$S = \int_1^4 2\pi\sqrt{x} \cdot \sqrt{\frac{4x+1}{4x}} dx = \int_1^4 2\pi\sqrt{x} \cdot \frac{\sqrt{4x+1}}{2\sqrt{x}} dx = \pi \int_1^4 \sqrt{4x+1} dx.$$

Let $u = 4x + 1$, $du = 4 dx$. Limits: $x = 1 \rightarrow u = 5$, $x = 4 \rightarrow u = 17$.

$$S = \frac{\pi}{4} \int_5^{17} \sqrt{u} du = \frac{\pi}{4} \cdot \frac{2}{3} [u^{3/2}]_5^{17} = \frac{\pi}{6} (17\sqrt{17} - 5\sqrt{5}) \approx 30.85.$$

Definition 11 (Surface Area about the y -Axis) If g is positive and has a continuous derivative on $[c, d]$, the area S of the surface generated by revolving the curve $x = g(y)$ about the y -axis is

$$S = \int_c^d 2\pi g(y) \sqrt{1 + [g'(y)]^2} dy.$$

Definition 12 (Surface Area for Parametric Curves) If a curve is given parametrically by $x = f(t)$, $y = g(t)$ for $t \in [\alpha, \beta]$, with $g(t) \geq 0$ and f' , g' continuous, the surface area generated by revolving about the x -axis is

$$S = \int_\alpha^\beta 2\pi g(t) \sqrt{[f'(t)]^2 + [g'(t)]^2} dt.$$

Example 9 Find the surface area of a sphere of radius r . The sphere is generated by revolving the semicircle $y = \sqrt{r^2 - x^2}$ for $x \in [-r, r]$ about the x -axis. $f'(x) = \frac{-x}{\sqrt{r^2 - x^2}}$, so

$$1 + [f'(x)]^2 = 1 + \frac{x^2}{r^2 - x^2} = \frac{r^2}{r^2 - x^2}.$$

Thus,

$$S = \int_{-r}^r 2\pi\sqrt{r^2 - x^2} \cdot \frac{r}{\sqrt{r^2 - x^2}} dx = 2\pi r \int_{-r}^r 1 dx = 2\pi r \cdot 2r = 4\pi r^2.$$

Definition 13 (General Surface Area Formula) For a curve revolved about a horizontal line $y = c$,

$$S = \int_a^b 2\pi |f(x) - c| \sqrt{1 + [f'(x)]^2} dx.$$

For a curve revolved about a vertical line $x = c$,

$$S = \int_c^d 2\pi |g(y) - c| \sqrt{1 + [g'(y)]^2} dy.$$

Example 10 Find the surface area generated by revolving $y = x^2$ on $[0, 1]$ about the y -axis. Using the parametric form with $x = t$, $y = t^2$ for $t \in [0, 1]$. Then $\frac{dx}{dt} = 1$, $\frac{dy}{dt} = 2t$, and the distance from the y -axis is $x = t$. Thus,

$$S = \int_0^1 2\pi t \sqrt{1 + 4t^2} dt.$$

Let $u = 1 + 4t^2$, $du = 8t dt$, so $t dt = \frac{du}{8}$. Limits: $t = 0 \rightarrow u = 1$, $t = 1 \rightarrow u = 5$.

$$S = 2\pi \cdot \frac{1}{8} \int_1^5 \sqrt{u} du = \frac{\pi}{4} \cdot \frac{2}{3} [u^{3/2}]_1^5 = \frac{\pi}{6} (5\sqrt{5} - 1) \approx 5.33.$$

Theorem 2 (Summary of Surface Area Formulas) • About x -axis: $S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$.

• About y -axis: $S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$.

• Parametric about x -axis: $S = \int_\alpha^\beta 2\pi y(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$.

• Parametric about y -axis: $S = \int_\alpha^\beta 2\pi x(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$.